

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Second Semester of M. Pharm (Pharmaceutical Technology)
University Theory Examination April/May 2018
PHPT005 NOVEL DRUG DELIVERY SYSTEMS-II

Date: 07/05/2018, Monday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 (A)** What is drug targeting? What are the levels of drug targeting? Differentiate between active and passive drug targeting. **05**
- (B)** Discuss the various factors affecting the nasal drug absorption. **05**
- Q.2 (A)** Explain the rationale behind targeting of drug to the brain using appropriate example of disease/disorder. **05**
- (B)** Write a note on anatomy and physiology of nose. **05**
- OR**
- Q.2 (A)** Explain with the help of labelled diagram, the anatomy of eye. **05**
- (B)** What are the different approaches for enhancing the drug absorption through nasal route? **05**
- Q.3 Answer the following (Any three) 15**
- (A)** Discuss the various applications of intranasal drug delivery.
- (B)** Using suitable example, explain the rationale for the formulation of targeted drug delivery systems.
- (C)** List down the novel formulation for ocular drug delivery. Explain in detail "Inserts".
- (D)** Write a note on dry powder inhalers.

SECTION-II

- Q.4** Write a note on various approaches for drug targeting to tumor cells. **10**
- Q.5** Explain pH triggered systems and Prodrug based systems for colon drug delivery with suitable examples. **10**
- OR**
- Q.5** Discuss the formulation and evaluation aspects of microparticulate based drug delivery systems for delivery to the lungs. **10**
- Q.6 Answer the following (Any three) 15**
- (A)** Explain invivo techniques for measurement of brain uptake.
- (B)** Describe the factors affecting the drug delivery to tumor cells.
- (C)** With a suitable diagram, explain the working of a Pulsincap® system employed for colon drug delivery.
- (D)** Discuss the biology and pharmacology of Blood Brain Barrier.